BLUESTOCK FINTECH

Mutual Fund Analytics Platform

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Project Scope: End-to-End ETL, DB, Analytics & Dashboard

1. Executive Summary

This capstone project presents the end-to-end development of the Bluestock Mutual Fund Analytics Platform. The platform automates data ingestion, cleans and validates transaction and performance schemas, stores transactional data in an optimized relational SQLite database, computes financial risk-return metrics including CAGR, Sharpe, Sortino, Alpha, Beta, VaR, CVaR, and Herfindahl concentration indexes, and delivers a premium interactive dashboard for investor decision making.

2. Problem Statement

Investors and advisors struggle with data fragmentation because NAV histories, transactions, and portfolio weights are distributed across separate AMFI files, web pages, and APIs in disparate formats. This project establishes a single unified SQLite data model, performs complex aggregations, and removes the reporting lag associated with static documentations.

3. ETL & Relational Database Design

The project follows a star-schema design optimizing dimensions and facts. A dynamically generated dim_date table allows uniform daily and monthly time-series groupings. Fact tables such as fact_nav and fact_transactions represent granular metrics, indexed to support sub-millisecond query performance.

4. Key Analytics Findings

• **Returns & Volatility:** Small Cap funds generated the highest 3-Year CAGR returns (>20%) but exhibited higher standard deviations and maximum drawdown risk.
• **Investor Cohorts:** Investors active since 2024 have shown a higher monthly transaction volume compared to new 2025 registrants, although the average SIP ticket size remained stable around Rs. 5,000.
• **SIP Continuity:** 12% of investors with active SIP plans were flagged as 'at-risk' due to payment gaps exceeding 35 days.

5. Recommendations & Limitations

Recommendations: Implement alerts for 'at-risk' SIP investors to prevent policy lapses. Integrate the automated fund scorecard directly inside client portfolio suggestion modules.
Limitations: This database relies on local historical extracts. Live API integrations are necessary to fetch expense ratios dynamically on a daily frequency.